Book Sharing Platform

Project Domain / Category
Web Programming

Abstract:
The Book Sharing Platform is a web-based community application that allows users to share and borrow both physical and digital books. Each registered user can act as both a sharer (when listing books) and a borrower (when requesting books). For **physical books**, the system not only tracks availability but also provides a simple communication mechanism for users to schedule offline exchanges (date, time, location). For **digital books**, the system supports uploads/downloads with logical "return" tracking. The project demonstrates essential full-stack concepts: user authentication, CRUD operations, request workflows, and database design using Flask and SQLite.

Functional Requirements:

✦ **User Management**

- Users can register, log in, and manage their profile.
- User roles:
- **Member (default):** Can share books and borrow books.
- **Admin:** Can manage all books and users, and remove inappropriate entries.

✦ **Book Management**

- Add new books with details: title, author, category, type (physical/digital).
- For **physical books**: include location/availability notes.
- For **digital books**: upload file (e.g., PDF) or provide a download link.
- Edit or delete only books added by the logged-in user.

✦ **Borrowing & Returning Workflow**

- Members can search/filter books by title, author, or category.
- **Borrow Request:**
- Borrower clicks *"Request to Borrow"*.
- A small form appears to propose **date, time, and location** for exchange.
- Request is stored in the database and sent to the owner's dashboard.
- **Owner Actions:**
- View incoming requests.
- Accept, reject, or suggest an alternative (edit request fields).
- **Borrow Confirmation:**
- Once accepted, system marks book as *Borrowed*.
- Borrow request details (date, time, location) are visible to both users.
- **Returning:**
- *Physical books:* Owner marks as *Returned*.
- *Digital books:* Borrower clicks *Mark as Returned* (or system auto-returns after 7 days).
- Book status always flows as: **Available → Borrowed → Returned**.

✦ **User Dashboard**

- **My Books (Shared):** Books added by the user + requests from others.

- **Borrowed Books:** Books borrowed by the user + scheduled exchange details.

✦ **Admin Dashboard**

1. View all listed books.
2. Remove duplicate/inappropriate entries.
3. Manage or block abusive users.

How Sharing & Borrowing Works:

Physical Books:

1. **Sharing:** Member adds details → marked *Available*.
2. **Borrowing:** Borrower submits request with proposed date, time, and location.
3. **Scheduling:** Owner reviews and accepts/rejects/suggests changes.
4. **Exchange:** Actual handover offline → system marks book as *Borrowed*.
5. **Returning:** Owner marks as *Returned* once the book is received back.

Digital Books:

1. **Sharing:** Member uploads file or adds download link.
2. **Borrowing:** Borrower clicks *Borrow* → download allowed, status *Borrowed*.
3. **Returning:** Borrower clicks *Mark as Returned* (or system auto-returns).

Required Tools

Frontend

1. HTML, CSS, JavaScript (UI, forms, and interactions)

Backend

1. Flask (Python) – backend framework
2. SQLite – database
3. Flask-Login – authentication/session management
4. Werkzeug – password hashing
5. Jinja2 – template rendering

Supervisor:

Name: Muhammad Ilyas

Email ID: Muhammad.ilyas@vu.edu.pk

MS Teams ID: ilyas.vu@outlook.com